

SASWAT K. NAYAK

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EDUCATION

Master of Science(Machine Learning in Materials), University of Wisconsin-Madison

Aug 2024

- Minor in **Computer Science**.

Bachelor in Chemical Engineering, Indian Institute of Technology Kharagpur

June, 2021

RELEVANT COURSES

Artificial Neural Networks, Software Engineering, Image Processing, Advanced Machine Learning, Deep Learning, Information Retrieval, Fluid Mechanics, Probability and Stochastic Processes, Regression and Time Series Analysis, Machine Learning in Materials, Algorithms and Data Structures, Linear Algebra, Graph Databases, Statistical Inference.

SKILLS

Technical Skills	Electrodynamics, Photonics, Javascript, Python, CI/CD, AWS, Docker, Gitlab, Pandas, R, Matplotlib, SQL, Tableau, GCP, R, Hive, MongoDB, Snowflake, Databricks, Multi-threading HTC, Agile, Scrum Master, Azure, Inferential methods, A/B testing, Maxwell Equations.
Machine Learning	NLP, Basic Computer Vision, Transformers, Tensorflow, PyTorch, GANs, Language Models. Hugging Face, MLOps, Spacy, Large Language Models(BERT, GPT-2, T5, RoBERTa), Big Data Engineering, ElasticSearch, XGBoost, Random Forest, Scikit-learn, Langchain

EXPERIENCE

Data Analyst(Full Time)

Jul 2022

Innovaccer Analytics(Healthcare)

- Enhanced query times by 40% by transitioning from MySQL to Neo4j. Led team in optimizing patient data extraction and update processes, reducing time by 90% and costs by 50% with NLP (NLTK, Spacy, RoBERTa) and auto-encoders.
- Led a team of interns to extract patient data from progress notes. Successfully optimized the patient data update process, achieving a 90% reduction in time required and a 50% decrease in associated costs by implementing NLP solutions(NLTK, Spacy, RoBERTa) and using auto-encoders for anomaly detection.
- Developed and implemented RESTful APIs to improve healthcare data operations. Applied DevOps(Docker, Kubernetes) practices, boosting deployment efficiency by 30% and fostering team collaboration.

Graduate Research Assistant: Machine Learning in Materials(Semiconductors)

Sept 2022 - Present 2023

UW Madison

Madison, WI

- Introduced novel explainable descriptors that significantly outperform traditional symmetry-based descriptors in machine learning tasks. These advancements achieved a notable 50% improvement in the effectiveness of identifying and categorizing photonic crystal structures, as substantiated by an F1-score enhancement from 48% to 94%.

Machine Learning Intern

Nov 2020 - Dec 2020

Ushur

- Designed and implemented an advanced algorithm combining Google Tesseract OCR with rule-based techniques to accurately extract customer data from insurance claim documents; securing victory in an internal hackathon.

Project Experience

- Engineered a high-performance filesystem backed by relational databases, featuring a user-friendly web interface for comprehensive CRUD operations. Spearheaded the development of an intuitive visualization tool, enabling users to effortlessly explore directories and subdirectories through a dynamic tree structure. Deployed on Vercel.
- Used LangChain and Llama2 to make my own custom chatbot which answers questions on my professional experience. Check it out here: [Chatbot](#)